

ADVANCED MACROECONOMICS I

HETEROGENEITY IN MACROECONOMICS: NUMERICAL METHODS AND APPLICATIONS

New York University, Spring 2026

Instructor:	Virgiliu Midrigan	Time:	Th 2:00 – 4:00 PM
Email:	vm50@nyu.edu	Place:	19 W 4th, 624

Objectives:

We study the role of firm and household heterogeneity in shaping macroeconomic aggregates and optimal policy interventions. We will focus on economies with incomplete markets, financial and product market imperfections and analyze the impact of redistributive policies. We will also analyze economies with non-convex adjustment costs. We will discuss extensively the numerical methods needed to solve these models and practice these by solving a number of increasingly complex model economies.

Course Requirements:

1. Problem Sets (75%)

I will assign weekly problem sets that will ask you to solve simple versions of the models we will study in class. I will supply a set of Matlab (or, when we discuss machine learning, Python) codes to get you started, and you will be expected to modify and extend these codes to solve the problems. If you prefer to use other languages, that is fine, but you will be on your own. For each problem set, I would like you to first solve the assigned problem, and then extend it in some way (e.g. by changing some of the assumptions of the model, or adding a new feature to it).

I would like you to submit a set of slides (10 pages at most, in PDF format) summarizing i) your results, ii) a discussion of the accuracy of your solution (e.g. tables and figures summarizing Euler or Bellman residuals and the time it took for the algorithms to converge, iii) a summary of any issues you encountered while solving the problem and how you addressed them, iv) a brief description of your extension, a motivation for it and your findings. Also be sure to summarize the key takeaways from your analysis and providing intuition for them.

You should submit these slides to me via email, by Wednesdays at midnight. Each student should submit their own slides, and work independently on the extension, but feel free to talk to each other.

If you get stuck, feel free to drop by my office to ask for help. The first few problem sets will be more challenging if you are not used to numerical computations, but over time you can recycle old code and get better at it. I am confident that AI can solve these problems for you, but I strongly encourage you to do the work yourself, at least initially, as this is the only way you will master the material. Once you get the hang of it, you can and should use AI to help automatize tedious tasks, such as copying tex code into your software of choice, make your code more efficient, and debug it. But the only way you can understand what AI gives you and determine if its proposed solution is correct is if you first practice on your own with simple problems, before you use AI to help you with more complex ones.

2. In-Class Presentations (25%)

We will spend the first 20 minutes of each class discussing the problem set due that week. Each week, two students will be assigned to present their slides to the class. I will use a random number generator to pick the two students, so everyone should be prepared to present each week. The presenters should also be prepared to answer questions from me and the other students in the class, especially about the intuition for their findings.

Course Outline:

I. Introduction

We briefly review the standard incomplete markets economy and outline the various topics we will introduce in this course.

II. Numerical Tools

We start by discussing the key computational tools we will use to solve the models in this course: solving non-linear equations, optimization, numerical integration and differentiation and functional approximation. I will also introduce some machine learning techniques that we will use later in the course to solve simple models.

- [1] Burkhard and Maussner. “Dynamic General Equilibrium Modeling.”
- [2] Miranda and Fackler. “Applied Computational Economics and Finance”.
- [3] Press et al. “Numerical Recipes: The Art of Scientific Computing.”
- [4] Sargent and Stachurski. QuantEcon.

III. Macroeconomic Implications of Household Credit Frictions

We discuss the aggregate consequences of shocks that tighten household credit.

- [1] Guerrieri, Lorenzoni, 2017. “Credit Crises, Precautionary Savings, and the Liquidity Trap,” *QJE*.
- [2] Werning, 2012. “Managing a Liquidity Trap.”
- [3] Jones Midrigan and Philippon, 2018. “Household Leverage and the Recession,” *Ecma*.
- [4] Kehoe, Midrigan, Pastorino, 2017. “Debt Constraints and Employment,” *JPE*.

IV. Aggregate Fluctuations in Heterogeneous Agent Economies

We discuss two approaches to computing the responses to aggregate shocks in economies with heterogeneity: the Krusell-Smith (1998) approach and the MIT shocks approach. To implement the latter, we compute transition dynamics non-linearly, as well as using the SSJ perturbation approach.

- [1] Krusell and Smith, 1997. “Income and Wealth Heterogeneity, Portfolio Choice, and Equilibrium Asset Returns,” *Macroeconomic Dynamics*.
- [2] Krusell and Smith, 1998. “Income and Wealth Heterogeneity in the Macroeconomy,” *JPE*.
- [3] Boppart, Krusell, Mitman, 2018. “Exploiting MIT shocks in heterogeneous-agent economies: the impulse response as a numerical derivative,” *JEDC*.
- [4] Auclert, Bardoczy, Rognlie and Straub, 2021. “Using the Sequence-Space Jacobian to Solve and Estimate Heterogeneous-Agent Models,” *Ecma*.

V. Income and Wealth Inequality and Redistribution

We discuss the impact of redistributive policies aimed at curbing inequality, as well as the evidence on rate of return heterogeneity that has been argued to generate wealth inequality.

- [1] Heathcote, Perri, Violante, 2020. “The Rise of US Earnings Inequality: Does the Cycle Drive the Trend?” *RED*.

- [2] Heathcote and Tsujiyama, 2021. “Optimal Income Taxation: Mirrlees Meets Ramsey,” *JPE*.
- [3] Boar and Midrigan, 2022. “Efficient Redistribution,” *JME*.
- [4] Bach, Calvet, Sodini, 2020. “Rich Pickings? Risk, Return and Skill in Household Wealth,” *AER*.
- [5] Fagereng, Guiso, Malacrino, Pistaferri, 2020. “Heterogeneity and Persistence in Returns to Wealth,” *Ecma*.

VI. Firm Dynamics and Misallocation

We discuss the workhorse model of firm dynamics and the impact of idiosyncratic distortions.

- [1] Hopenhayn and Rogerson, 1993. “Job Turnover and Policy Evaluation: A General Equilibrium Analysis,” *JPE*.
- [2] Restuccia and Rogerson, 2008. “Policy Distortions and Aggregate Productivity with Heterogeneous Establishments,” *RED*.
- [3] Guner, Ventura, Xu, 2008. “Macroeconomic Implications of Size-Dependent Policies,” *RED*.
- [4] Hsieh and Klenow, 2009. “Misallocation and Manufacturing TFP in China and India,” *QJE*.
- [5] Hopenhayn, 2014. “On the Measure of Distortions,” *NBER WP 20404*.

VII. Firm-Level Credit Frictions and Misallocation

We discuss the impact of firm credit constraints on misallocation.

- [1] Buera, Kaboski, Shin, 2009. “Finance and Development: A Tale of Two Sectors,” *AER*.
- [2] Midrigan and Xu, 2013. “Finance and Misallocation: Evidence from Plant-Level Data,” *AER*.
- [3] Moll, 2014. “Productivity Losses from Financial Frictions: Can Self-financing Undo Capital Misallocation?” *AER*.
- [4] Boar, Midrigan and Gorea. “Why Are Returns to Wealth So Dispersed?” *NBER WP 29705*.

VIII. Markups

We discuss the measurement of markups and several models of endogenously variable markups.

- [1] Edmond, Midrigan and Xu, 2015. “Competition, Markups and the Gains from International Trade,” *AER*.
- [2] Edmond, Midrigan and Xu, 2018. “How Costly Are Markups?” *JPE*.
- [3] Boar and Midrigan 2019. “Markups and Inequality,” *REStud*.

IX. Economies with Non-Convex Adjustment Costs

We discuss the aggregate implications of economies in which firms face menu costs of price adjustment.

- [1] Stokey, 2009. “The Economics of Inaction,” Princeton University Press.
- [2] Golosov and Lucas, 2007. “Menu Costs and Phillips Curves,” *JPE*.
- [3] Midrigan, 2010. “Menu Costs, Multi-Product Firms and Aggregate Fluctuations,” *Ecma*.
- [4] Alvarez and Lippi, 2013. “Price Setting with Menu Costs for Multi-Product Firms,” *Ecma*.
- [5] Alvarez, Lippi and Le Bihan, 2014. “The real effects of monetary shocks in sticky price models: a sufficient statistic approach,” *AER*.

- [6] Auclert, Rigato, Rognlie, Straub, 2022. “New Pricing Models, Same Old Phillips Curves?”
- [7] Blanco, Boar, Jones, Midrigan, 2024. “Nonlinear Inflation Dynamics in Menu Cost Economies.”
- [8] Blanco, Boar, Jones, Midrigan, 2024. “The Inflation Accelerator”
- [9] Khan and Thomas, 2008. “Idiosyncratic Shocks and the Role of Nonconvexities in Plant and Aggregate Dynamics,” *Econometrica*.